The atmosphere

Earth is surrounded by a blanket of air called the atmosphere. This acts as a sunscreen by day, protecting us from the Sun's ultraviolet radiation. At night, it retains heat, keeping us warm. Air currents in the lowest level of the atmosphere carry heat around the globe, allowing life to flourish almost everywhere.

LAYERS OF AIR

The atmosphere is made up of a series of layers. These are formed by temperature changes that stop the air from moving from one layer to another. The lowest layer is called the troposphere. Above that lies the stratosphere, containing the ozone that shields us from harmful ultraviolet radiation. Above that are the mesosphere and the thermosphere.

ATMOSPHERIC GASES

Viewed from space, our atmosphere forms a glowing blue envelope around the planet. About three-quarters of it is made up of nitrogen. Most of the rest is oxygen. The air also contains argon and a small amount of carbon dioxide. Other gases include neon, helium, methane, krypton, hydrogen, nitrous oxide, and xenon, along with water vapor and ozone.

Thermosphere
The outermost, and deepest, layer starts at around 50 miles (80 km) and extends up to 300 miles (500 km), fading into the vacuum of space.

Mesosphere
The top of this layer, which extends from 30 to 50 miles (50–80 km), is the coldest part of the atmosphere, with temperatures plunging to -112°F (-80°C).

Stratosphere
Extending from 10 to 30 miles (16–50 km), this layer gets hotter with altitude, unlike the troposphere where the opposite happens.

Troposphere
The lowest layer is also the shallowest, up to 10 miles (16 km) deep.